

Importing Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
import warnings
warnings.filterwarnings('ignore')
```

Load Data

```
data = pd.read_csv("Obesity dataset.csv")#load data
```

Data Cleaning

```
data.head()
```

	Gender	Age	Height	Weight	family_history	FAVC	FCVC	NCP	CAEC	SMOKE	CH20	SCC	FAF	TUE	CALC	
0	Male	39.2	1.49	79.2	no	yes	1	1	Always	no	2	no	0	2	Frequently	A
1	Male	55.2	1.74	150.5	yes	no	3	1	Frequently	no	2	no	0	1	Frequently	
2	Male	15.6	1.61	143.6	yes	yes	3	1	Sometimes	no	1	no	1	2	Sometimes	A
3	Female	23.3	1.70	160.8	yes	yes	2	2	Frequently	no	2	no	0	2	Frequently	
4	Female	59.3	1.87	60.9	no	yes	2	4	no	no	1	no	3	1	Sometimes	Public_Tran

Next steps: [Generate code with data](#) [New interactive sheet](#)

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2111 entries, 0 to 2110
Data columns (total 17 columns):
#   Column              Non-Null Count  Dtype
---  -
0   Gender              2111 non-null   object
1   Age                 2111 non-null   float64
2   Height              2111 non-null   float64
3   Weight              2111 non-null   float64
4   family_history      2111 non-null   object
5   FAVC                2111 non-null   object
6   FCVC                2111 non-null   int64
7   NCP                 2111 non-null   int64
8   CAEC                2111 non-null   object
9   SMOKE               2111 non-null   object
10  CH20                2111 non-null   int64
11  SCC                 2111 non-null   object
12  FAF                 2111 non-null   int64
13  TUE                 2111 non-null   int64
14  CALC                2111 non-null   object
15  MTRANS              2111 non-null   object
16  Obesity              2111 non-null   object
dtypes: float64(3), int64(5), object(9)
memory usage: 280.5+ KB
```

There are total 2111 entries. 17 columns,2111 rows. Age is in float but it should be in integer datatype.

```
# Checking the missing values
data.isnull().sum()
```

```
      0
Gender  0
Age     0
Height  0
Weight  0
family_history  0
FAVC     0
FCVC     0
NCP      0
CAEC     0
SMOKE    0
CH2O     0
SCC      0
FAF      0
TUE      0
CALC     0
MTRANS   0
Obesity  0
```

dtype: int64

As we can see, there are no null values in any of the columns

Age can't be like 21.0,. So,Converting Age column float datatype into integer datatype.

```
data['Age'] = data['Age'].apply(np.int64)
```

```
data.dtypes
```

```
      0
Gender  object
Age     int64
Height  float64
Weight  float64
family_history  object
FAVC     object
FCVC     int64
NCP      int64
CAEC     object
SMOKE    object
CH2O     int64
SCC      object
FAF      int64
TUE      int64
CALC     object
MTRANS   object
Obesity  object
```

dtype: object

```
data.head()
```

	Gender	Age	Height	Weight	family_history	FAVC	FCVC	NCP	CAEC	SMOKE	CH20	SCC	FAF	TUE	CALC	
0	Male	39	1.49	79.2	no	yes	1	1	Always	no	2	no	0	2	Frequently	Aut
1	Male	55	1.74	150.5	yes	no	3	1	Frequently	no	2	no	0	1	Frequently	
2	Male	15	1.61	143.6	yes	yes	3	1	Sometimes	no	1	no	1	2	Sometimes	Aut
3	Female	23	1.70	160.8	yes	yes	2	2	Frequently	no	2	no	0	2	Frequently	
4	Female	59	1.87	60.9	no	yes	2	4	no	no	1	no	3	1	Sometimes	Public_Trans

Next steps:

[Generate code with data](#)[New interactive sheet](#)

Obesity level column is in categorical form & it need to be converted into numeric.

```
#Converting Obesity level column in numerical using encoding technique.
import pandas as pd
from sklearn.preprocessing import LabelEncoder

label_encoder = LabelEncoder()
# Fit and transform the 'Obesity' column
data['obesity_level_encoded'] = label_encoder.fit_transform(data['Obesity'])
```

Checking Duplicates

```
# Check for duplicate rows
duplicate_rows = data[data.duplicated()]
print("Duplicate Rows:")
print(duplicate_rows)
```

```
864  Female  46  1.90  112.6  no  no  2  4  no
904  Male  28  1.84  166.8  yes no  3  2  Sometimes
920  Male  18  1.97  92.7  no  no  1  4  Frequently
1032 Female  47  1.57  124.8  no  yes  2  2  Always
1065 Male  55  1.80  94.5  yes  yes  1  4  Sometimes
1140 Female  31  1.54  57.7  no  no  3  2  Frequently
1448 Female  32  1.69  71.1  no  no  3  3  no
1577 Female  55  1.56  139.5  yes no  1  2  Sometimes
1593 Male  45  1.90  39.0  no  yes  2  4  Always
1641 Male  56  1.67  39.4  no  no  3  3  Always
1706 Male  51  1.61  172.8  yes  yes  1  1  Always
1733 Male  56  1.96  43.2  no  no  3  1  no
1761 Female  29  1.60  43.9  no  no  1  1  Always
```

```
SMOKE  CH20  SCC  FAF  TUE  CALC  MTRANS \
69  no  3  no  2  1  Sometimes  Motorbike
111  yes  2  no  0  1  Sometimes  Bike
220  no  3  no  0  2  Frequently  Walking
262  no  1  yes  2  2  Frequently  Walking
397  yes  2  no  1  2  Frequently  Automobile
435  no  2  no  2  0  no  Public_Transportation
606  yes  1  no  0  1  Sometimes  Public_Transportation
671  no  2  no  2  1  Frequently  Motorbike
691  no  2  no  2  1  Sometimes  Automobile
809  no  1  no  1  0  Frequently  Motorbike
826  yes  1  no  0  0  Sometimes  Motorbike
864  no  2  no  0  0  Sometimes  Public_Transportation
904  yes  2  no  1  0  Sometimes  Automobile
920  yes  3  yes  3  0  Sometimes  Bike
1032  yes  3  yes  2  0  Frequently  Motorbike
1065  yes  2  yes  0  1  Sometimes  Walking
1140  no  3  no  1  0  Sometimes  Motorbike
1448  yes  1  yes  1  0  no  Automobile
1577  yes  3  no  2  1  Frequently  Automobile
1593  no  2  no  3  1  Sometimes  Public_Transportation
1641  no  2  no  3  1  Frequently  Motorbike
1706  yes  3  no  2  2  Frequently  Public_Transportation
1733  no  2  yes  0  1  Frequently  Walking
1761  no  3  no  1  0  no  Motorbike
```

```
904 Obesity_Type_I 2
920 Normal_Weight 1
1032 Overweight_Level_I 5
1065 Overweight_Level_II 6
1140 Obesity_Type_II 3
```

Removing duplicates

```
data_cleaning = data.drop_duplicates()
```

Earlier we had 2111 rows and now we have 2087 rows i.e. 24 duplicate rows were there which should be removed.

```
data.head()
```

	Gender	Age	Height	Weight	family_history	FAVC	FCVC	NCP	CAEC	SMOKE	CH2O	SCC	FAF	TUE	CALC	
0	Male	39	1.49	79.2	no	yes	1	1	Always	no	2	no	0	2	Frequently	Au
1	Male	55	1.74	150.5	yes	no	3	1	Frequently	no	2	no	0	1	Frequently	
2	Male	15	1.61	143.6	yes	yes	3	1	Sometimes	no	1	no	1	2	Sometimes	Au
3	Female	23	1.70	160.8	yes	yes	2	2	Frequently	no	2	no	0	2	Frequently	
4	Female	59	1.87	60.9	no	yes	2	4	no	no	1	no	3	1	Sometimes	Public_Trans

Next steps:

[Generate code with data](#)

[New interactive sheet](#)

```
data.dtypes
```

```

      0
Gender      object
Age         int64
Height      float64
Weight      float64
family_history  object
FAVC        object
FCVC        int64
NCP         int64
CAEC        object
SMOKE       object
CH2O        int64
SCC         object
FAF         int64
TUE         int64
CALC        object
MTRANS      object
Obesity     object
obesity_level_encoded  int64
```

dtype: object

```
data['Weight'] = data['Weight'].apply(np.int64)
```

```
data.dtypes
```

```

0
Gender      object
Age         int64
Height      float64
Weight      int64
family_history  object
FAVC        object
FCVC        int64
NCP         int64
CAEC        object
SMOKE       object
CH2O        int64
SCC         object
FAF         int64
TUE         int64
CALC        object
MTRANS      object
Obesity     object
obesity_level_encoded  int64

```

dtype: object

data.head()

	Gender	Age	Height	Weight	family_history	FAVC	FCVC	NCP	CAEC	SMOKE	CH2O	SCC	FAF	TUE	CALC	
0	Male	39	1.49	79	no	yes	1	1	Always	no	2	no	0	2	Frequently	Au
1	Male	55	1.74	150	yes	no	3	1	Frequently	no	2	no	0	1	Frequently	
2	Male	15	1.61	143	yes	yes	3	1	Sometimes	no	1	no	1	2	Sometimes	Au
3	Female	23	1.70	160	yes	yes	2	2	Frequently	no	2	no	0	2	Frequently	
4	Female	59	1.87	60	no	yes	2	4	no	no	1	no	3	1	Sometimes	Public_Trans

Next steps: [Generate code with data](#) [New interactive sheet](#)

Done with data cleaning.

Columns: Gender, Age, Height, Weight, family_history, FAVC: Frequent consumption of high-calorie food, FCVC: Frequency of consumption of vegetables, NCP: Nutrition Care Process, No. of main meals, CAEC: Consumption of food between meals, Smoke, CH2O Consumption of daily water, SCC: Somatic Cell Count, Calories Consumption, FAF: Physical Activity Frequency, TUE: Time using technology devices, CALC: Alcohol Consumption, MTRANS: Means of Transportation, Obesity, obesity_level_encoded

For better visualization Creating another column to define Age groups.

```

def categorize_age(age):
    if age >= 60:
        return 'Senior'
    elif age >= 20:
        return 'Adult'
    elif age >= 13:
        return 'Teenager'
    else:
        return 'Child'

# create Age Group column
data['Age Group'] = data['Age'].apply(categorize_age)

```

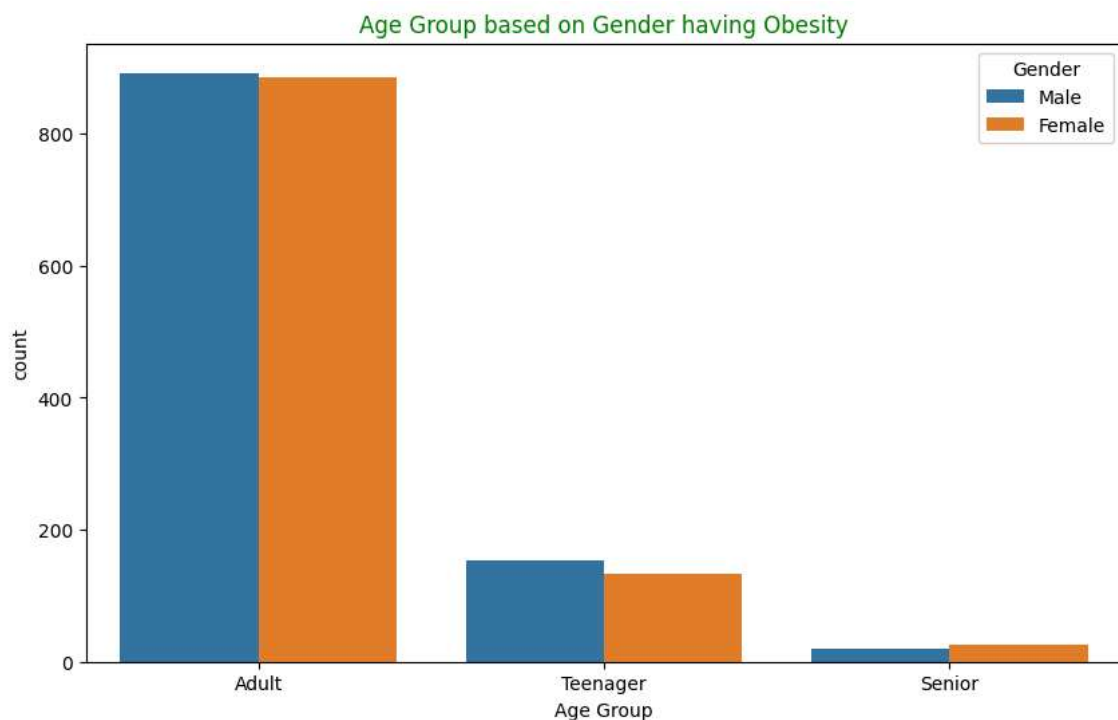
data.head()

	Gender	Age	Height	Weight	family_history	FAVC	FCVC	NCP	CAEC	SMOKE	CH20	SCC	FAF	TUE	CALC	
0	Male	39	1.49	79	no	yes	1	1	Always	no	2	no	0	2	Frequently	Ad
1	Male	55	1.74	150	yes	no	3	1	Frequently	no	2	no	0	1	Frequently	
2	Male	15	1.61	143	yes	yes	3	1	Sometimes	no	1	no	1	2	Sometimes	Ad
3	Female	23	1.70	160	yes	yes	2	2	Frequently	no	2	no	0	2	Frequently	

Next steps: [Generate code with data](#) [New interactive sheet](#)

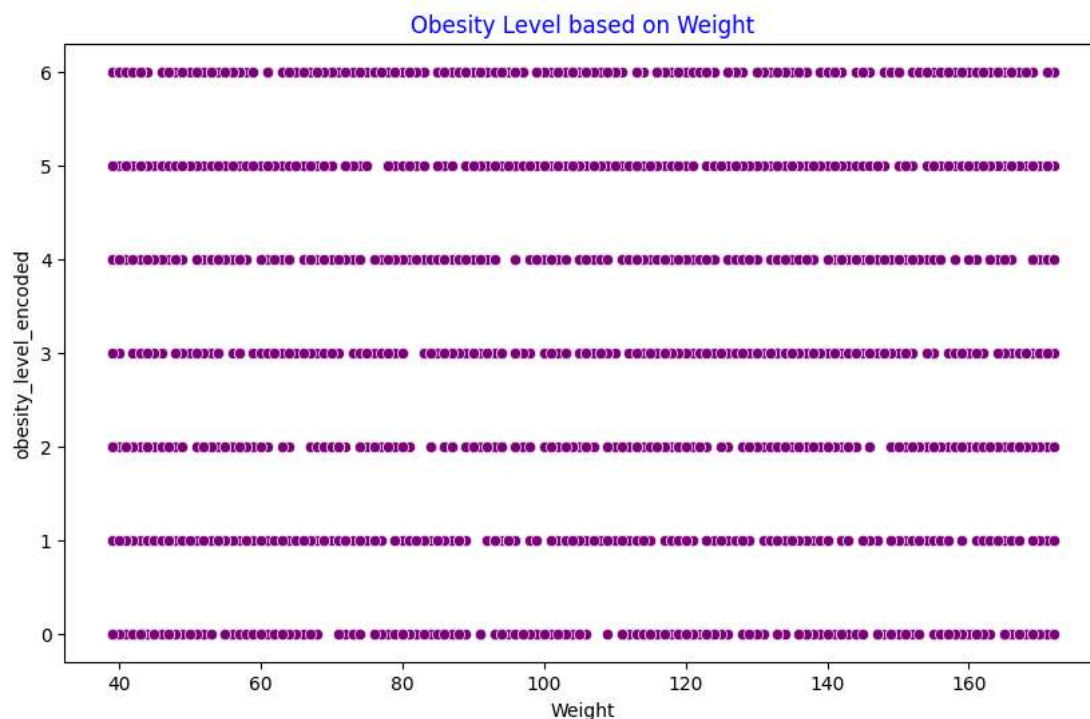
Now, We can compare the Gender based on Age groups having Obesity

```
# Age Group based on Gender using bar plot
plt.figure(figsize=(10,6))
sns.countplot(data=data,x='Age Group',hue='Gender')
plt.title('Age Group based on Gender having Obesity',color='g')
plt.show()
```



Here we can see, 1. Adults are more suffering from this disease. 2. Adult males are more suffering than females. 3. Small ratio of teenagers. 4. No more seniors suffering. 5. Only 1 female from senior age group.

```
# Obesity Level based on Weight using scatter plot
plt.figure(figsize=(10,6))
sns.scatterplot(data=data,x='Weight',y='obesity_level_encoded',color='purple')
plt.title('Obesity Level based on Weight',color='b')
plt.show()
```

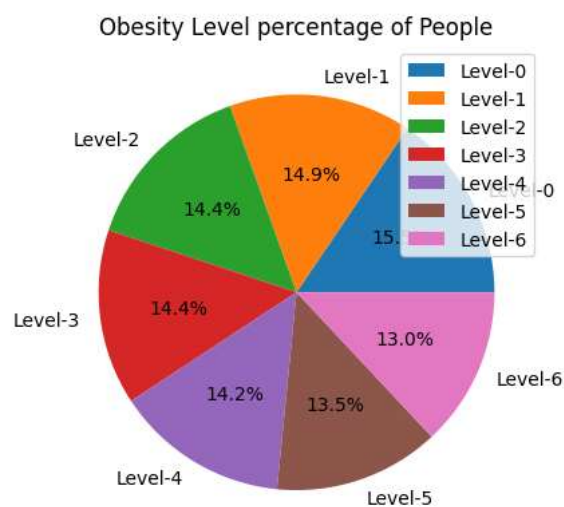


- Here we can see that,
1. Obesity levels are represented as discrete categories (0–6), resulting in horizontal bands across the plot.
 2. Weight values are distributed across all obesity categories, ranging approximately from 40 kg to 170 kg.
 3. There is no strong visible trend or correlation between weight and obesity level in this visualization alone, as similar weight values appear across multiple obesity categories.
 4. The overlap of points suggests that weight by itself is insufficient to determine obesity level accurately.

From the above we can see that,

1. Males are consuming more alcohol than females.
2. Only one male is daily consuming alcohol, no other entries of daily consuming alcohol.
3. No males are there in obesity level-4 and only few patients belongs to Obesity level-4.

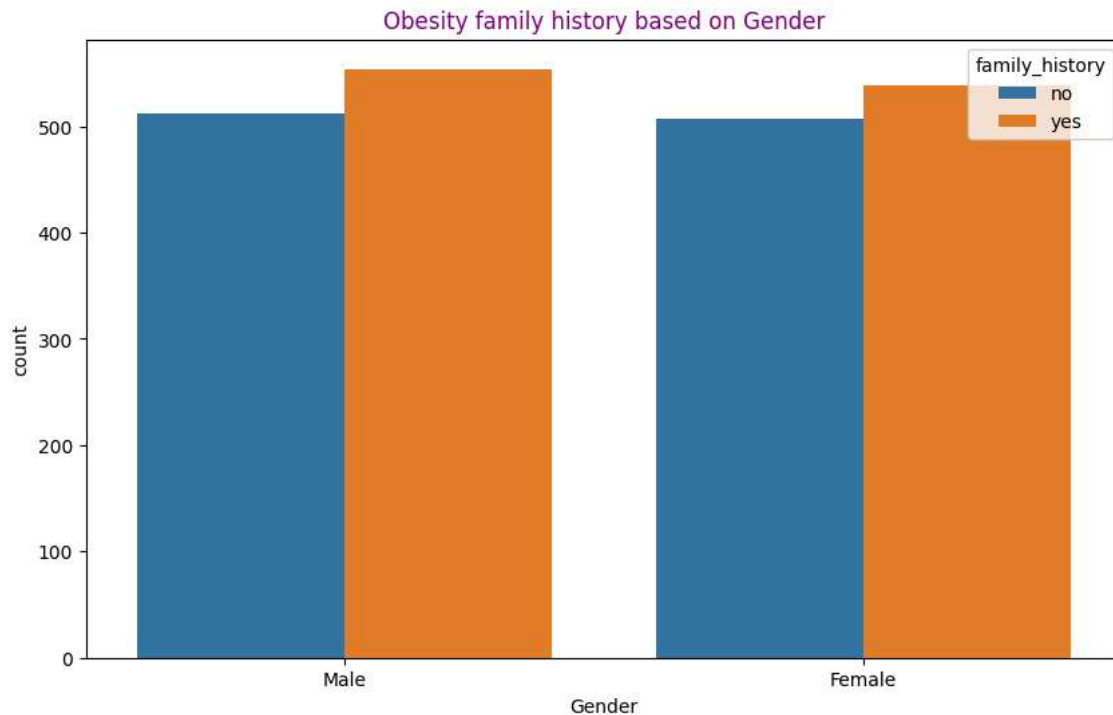
```
#plot pie chart based on obesity_level_encoded
obesity_counts = data['obesity_level_encoded'].value_counts()
labels=['Level-0','Level-1','Level-2','Level-3','Level-4','Level-5','Level-6',]
# Plot the pie chart
plt.pie(obesity_counts, labels=labels, autopct='%1.1f%%')
plt.title('Obesity Level percentage of People')
plt.legend()
plt.show()
```



From above we can see that,

1. Level-0 people are more i.e. highest 15% people belong.
2. Level-6 people are less i.e. lowest only 13% people belong.
3. There are equal percentage of people belong to Level-2 & 3 i.e. 14.4% people.

```
# Obesity family history based on Gender using bar plot
plt.figure(figsize=(10,6))
sns.countplot(data=data,x='Gender',hue='family_history')
plt.title('Obesity family history based on Gender',color='purple')
plt.show()
```



From the above, 1. There are more male than females who are suffering from the obesity. 2. The ratio of family members having obesity of both male and females are high. So, we can consider as most of the people are having obesity because of genetics.

Summary

Adults are suffering more from these disease . Adult males are suffering more than females . Teenagers ratio is medium. Seniors are healthy as only few are having this disease. People having weight upto 70 are normal. People having weight between 50-110 are having highest obesity level. More people are in the starting phase of obesity & very less people are suffering from highest level of obesity. More ratio of family members having obesity i.e. It may happening because of genetics.

```
!pip install ydata-profiling
```

```
Collecting ydata-profiling
  Downloading ydata_profiling-4.18.4-py2.py3-none-any.whl.metadata (23 kB)
Requirement already satisfied: scipy<1.17,>=1.8 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (1.16.3)
Requirement already satisfied: pandas!=1.4.0,<3.0,>=1.5 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (
Requirement already satisfied: matplotlib<=3.10,>=3.5 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (3
Requirement already satisfied: pydantic<3,>=2 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (2.12.3)
Requirement already satisfied: PyYAML<6.1,>=6.0.3 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (6.0.3
Requirement already satisfied: Jinja2<3.2,>=3.1.6 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (3.1.6
Collecting visions<0.8.2,>=0.7.5 (from visions[type_image_path]<0.8.2,>=0.7.5->ydata-profiling)
  Downloading visions-0.8.1-py3-none-any.whl.metadata (11 kB)
Requirement already satisfied: numpy<2.4,>=1.22 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (2.0.2)
Collecting minify-html<=0.15.0 (from ydata-profiling)
  Downloading minify_html-0.18.1-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (18 kB)
Collecting filetype>=1.0.0 (from ydata-profiling)
  Downloading filetype-1.2.0-py2.py3-none-any.whl.metadata (6.5 kB)
Collecting phik<0.13,>=0.12.5 (from ydata-profiling)
  Downloading phik-0.12.5-cp312-cp312-manylinux_2_24_x86_64.manylinux2014_x86_64.whl.metadata (5.6 kB)
Requirement already satisfied: requests<3,>=2.32.0 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (2.32
Requirement already satisfied: tqdm<5,>=4.66.3 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (4.67.3)
Requirement already satisfied: seaborn<0.14,>=0.10.1 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (0.
Collecting multimethod<2,>=1.4 (from ydata-profiling)
  Downloading multimethod-1.12-py3-none-any.whl.metadata (9.6 kB)
Requirement already satisfied: statsmodels<1,>=0.13.2 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (0
Requirement already satisfied: typeguard<5,>=4 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (4.5.2)
Collecting imagehash==4.3.2 (from ydata-profiling)
  Downloading ImageHash-4.3.2-py2.py3-none-any.whl.metadata (8.4 kB)
Requirement already satisfied: wordcloud>=1.9.4 in /usr/local/lib/python3.12/dist-packages (from ydata-profiling) (1.9.6)
Collecting dacite<2,>=1.9 (from ydata-profiling)
```